# Eicosanoids

**Before: CPR lecture: Primary and secondary hemostasis**

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https://www.youtube.com/watch?v=-JOn8g8LvwE

# DLA: Eicosanoids

| | |
|---|---|
| SOM.MK.I.BPM1.1.CPR.2.BCHM. 0252 | Describe the synthesis of the eicosanoid molecules, their general structure, and their precursors. Indicate inhibitors of eicosanoid synthesis. |
| SOM.MK.I.BPM1.1.CPR.2.BCHM. 0253 | Discuss the role of eicosanoids in the regulation of inflammation (PGE2), gastric acidity, blood flow, and platelet aggregation (PGI2 and TXA2). |
| SOM.MK.I.BPM1.1.CPR.2.BCHM. 0254 | Compare and contrast the effects of TXA2 vs TXA3 and their effects on vascular health. |
| SOM.MK.I.BPM1.1.CPR.2.BCHM. 0255 | Discuss the role of leukotrienes, cysteinyl leukotrienes and SRS-A in asthma and in anaphylactic shock |

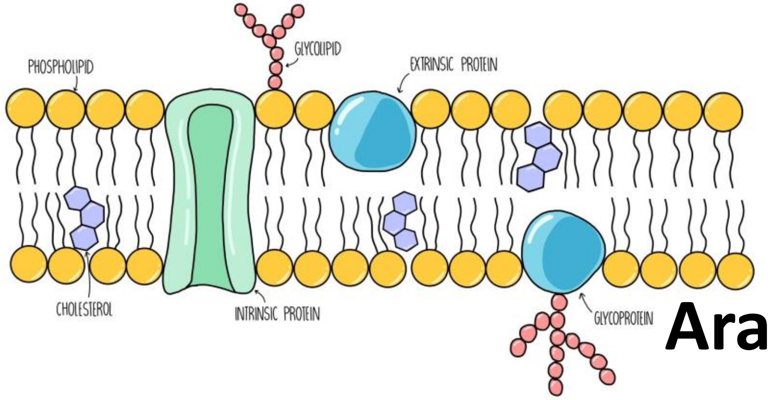
# Recommended Reading

- Chapter 17 (pg 236-239) in Lippincott's Illustrated Reviews of Biochemistry: Phospholipid, Glycosphingolipid, and Eicosanoid Metabolism | Lippincott® Illustrated Reviews: Biochemistry, 8e | Medical Education | Health Library (lwwhealthlibrary.com): Section on Eicosanoid metabolism
  - Questions: 17.1,17.5
- Practice questions on Sakai
- Lipid-Derived Autacoids: Eicosanoids and Platelet-Activating Factor | Goodman & Gilman's: The Pharmacological Basis of Therapeutics, 14e | AccessMedicine | McGraw Hill Medical (oclc.org)
- The Eicosanoids: Prostaglandins, Thromboxanes, Leukotrienes, & Related Compounds | Basic & Clinical Pharmacology, 15e | AccessMedicine | McGraw Hill Medical (oclc.org)
- Prostaglandins & Other Eicosanoids | Katzung & Trevor's Pharmacology: Examination & Board Review, 13e | AccessMedicine | McGraw Hill Medical (oclc.org)

# Eicosanoids

- Derived from 20-C (eicosa) fatty acids
- Lipid-derived mediators
- Synthesized by a variety of tissues
- Generally, have local effects (autocoids)
- Synthesized in small amounts; Short half-life
- Biologic effects mediated by binding to G-protein coupled receptors which result in intracellular changes in cAMP levels or intracellular $Ca^{2+}$ levels (Gs, Gi, Gq)

# Eicosanoid synthesis




**Membrane Phospholipid**

GLYCEROL — Saturated Fatty acid$_1$

GLYCEROL — Unsaturated Fatty acid$_2$

GLYCEROL — Phosphate — Nitrogen containing base

↓ **Phospholipase $A_2$**

**Arachidonic acid (20C-fatty acid)**

**Cyclooxygenase** → PGG2 → PGE2, PGF2$\alpha$, PGI2, TXA2

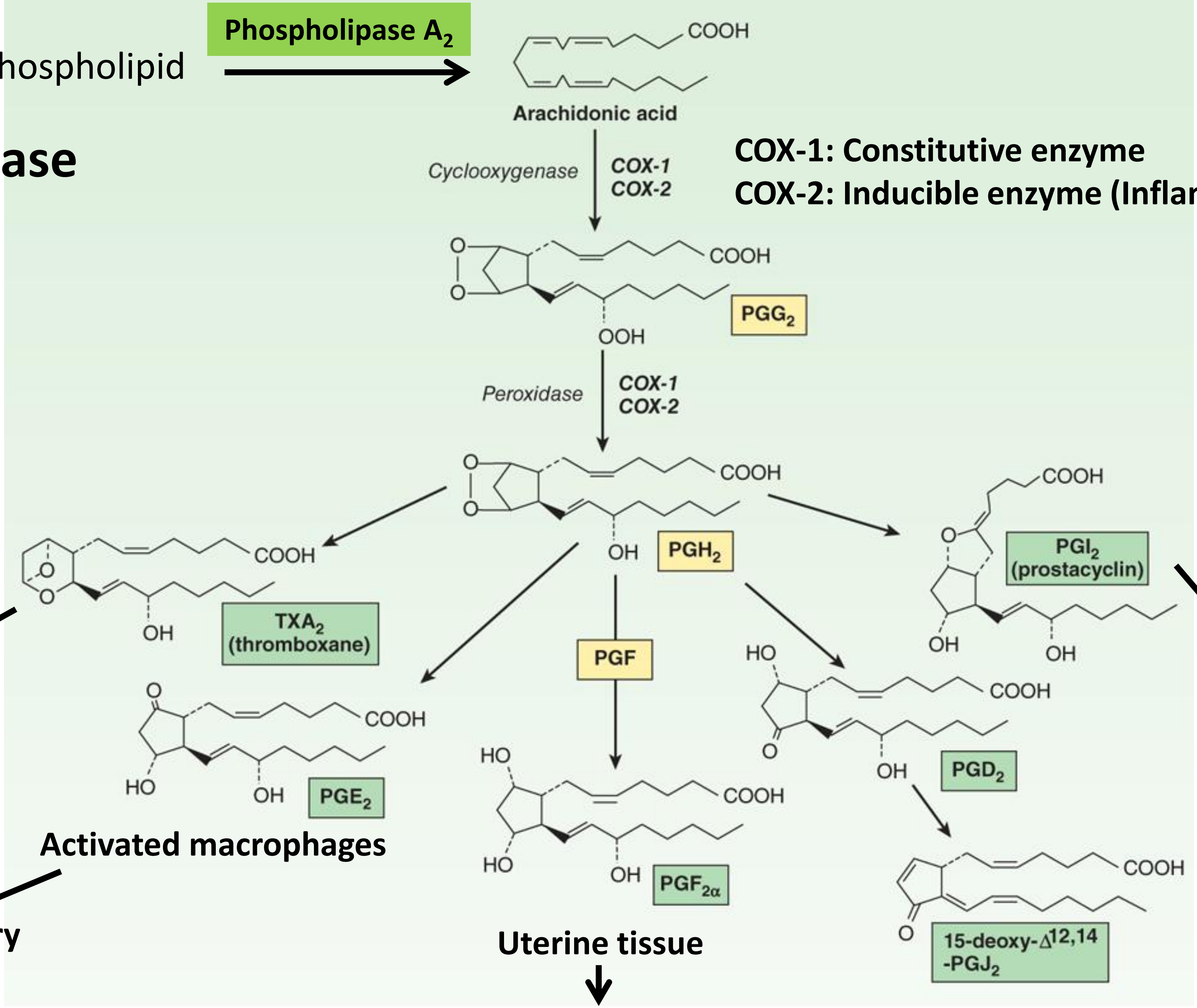
**Lipoxygenase** → Leukotrienes

# Prostanoids (Prostaglandin and Thromboxane)

- Formed from Arachidonic acid by **Cyclooxygenase (COX)**
  - **COX-1: Constitutive enzyme**
  - **COX-2: Inducible enzyme (Inflammation)**
- Produced by variety of tissues in a cell-specific manner
- Type of prostanoid produced depends on
  - Tissue and stimulus
  - Substrate lipid (Arachidonic acid)
- PGE2 – Activated macrophages
- PGF2$\alpha$ - Uterine tissue
- PGI2 (Prostacyclin) – Endothelium of blood vessels
- TXA2 (Thromboxane A2) – Activated platelets

# Cyclooxygenase pathway

Membrane Phospholipid




COX-1: Constitutive enzyme

COX-2: Inducible enzyme (Inflammation)

Platelets

Facilitate platelet aggregation

Activated macrophages

Mediate inflammatory response

Uterine tissue

Facilitate uterine contraction

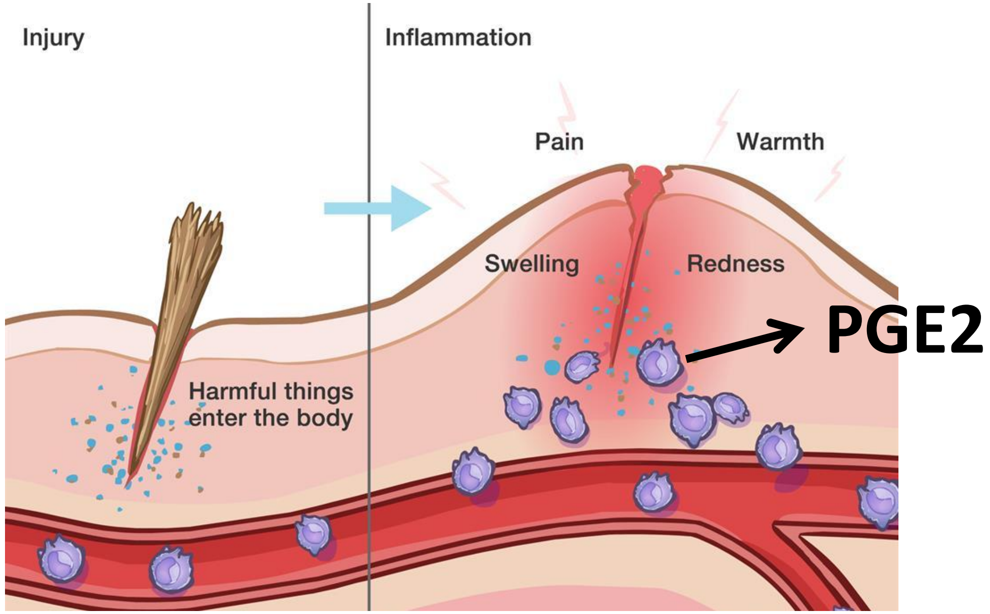
Endothelium

Inhibit platelet aggregation

Source: Bertram G. Katzung:
Basic & Clinical Pharmacology, Fourteenth Edition

# Biologic activities of PGE2 and PGF2$\alpha$

- **PGE2**: Mediator of inflammation; Released from activated macrophages;
  - Cause local **vasodilation** (redness); pain; fever (Signs of inflammatory response)
  - Protect gastric mucosa
- PGE2 and PGF2$\alpha$ : Uterine tissue: Induction of labor in pregnant uterus




**Mediator of inflammation**

# Biologic activities of PGI2 and TXA2

- **PGI2 (Prostacyclin):**
  - Endothelium → **Inhibits platelet aggregation**
  - **Vasodilation**

- **TXA2 (Thromboxane A2):**
  - Activated platelets →**Platelet aggregation**
  - **Vasoconstriction**
- Note that PGI2 and TXA2 are antagonists
- Ratio of PGI2 : TXA2 determines platelet aggregation

**PGI2 (Prostacyclin)**

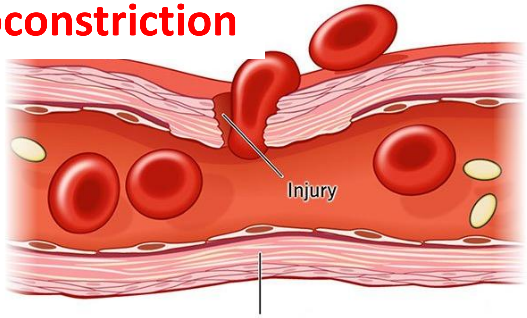
Vasodilation

Inhibits platelet aggregation

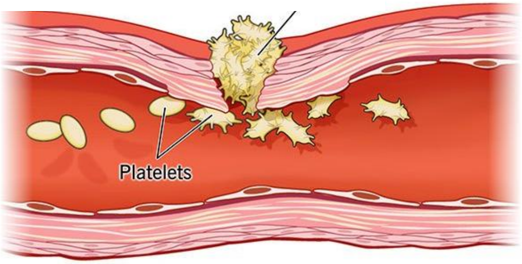
**TXA2 (Thromboxane A2)**

Vasoconstriction

Vasoconstriction




Platelet aggregation

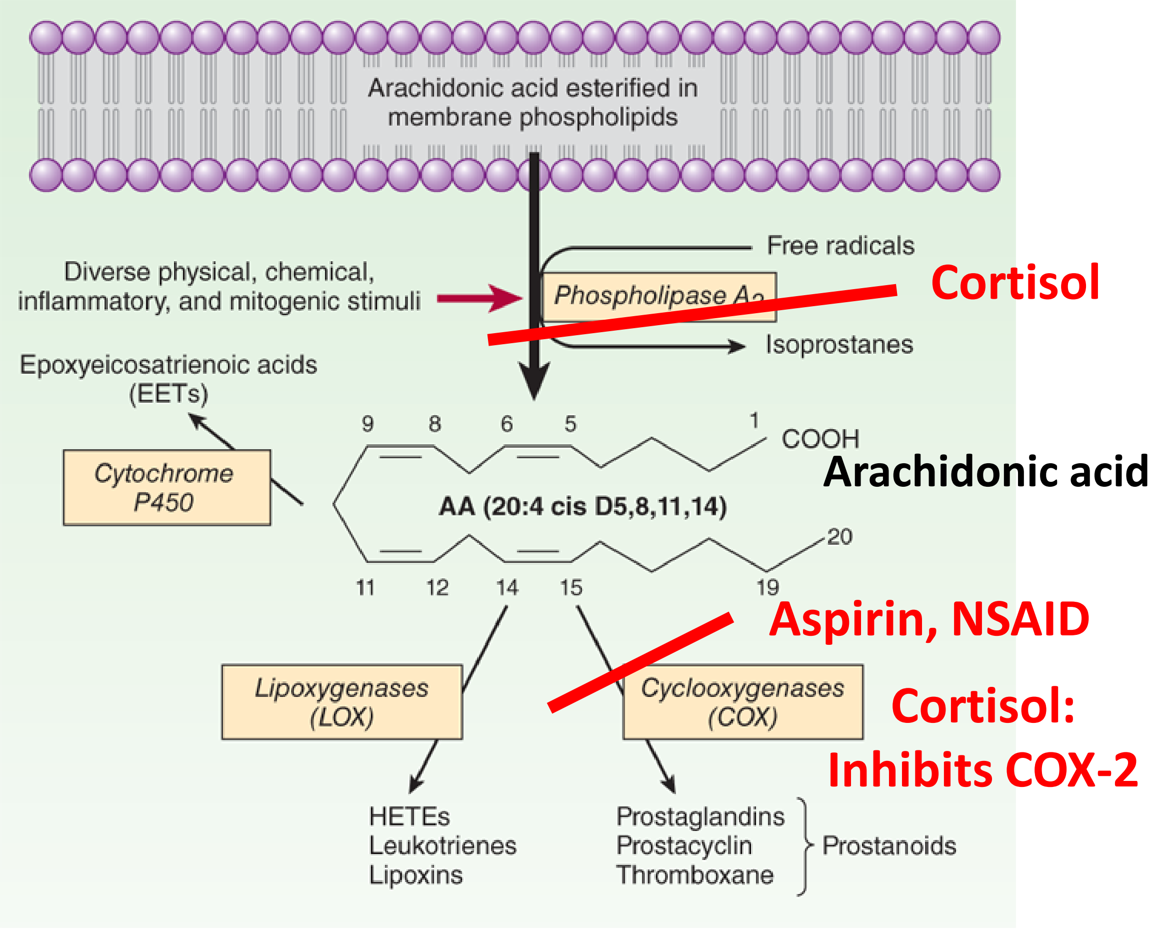


# Series 2 vs Series 3 Prostanoids

- Arachidonic acid (20C-fatty acid; four double bonds; $\omega$-6 fatty acid)
  - Linoleic acid (essential $\omega$-6 fatty acid precursor)
  - Precursor of **series 2 prostanoids (**PGI2, **TXA2)**
- Eicosapentaenoic acid (20C-fatty acid; 5 double bonds; $\omega$-3 fatty acid)
  - Precursor of **series 3 prostanoids (TXA3)**
  - Alpha-Linolenic acid (essential $\omega$-3 fatty acid precursor)
- **TXA3** less thrombogenic than **TXA2**
- Increasing dietary $\omega$-3 fatty acids increases **TXA3** and decreases risk of platelet aggregation

# Inhibitors of prostanoid synthesis

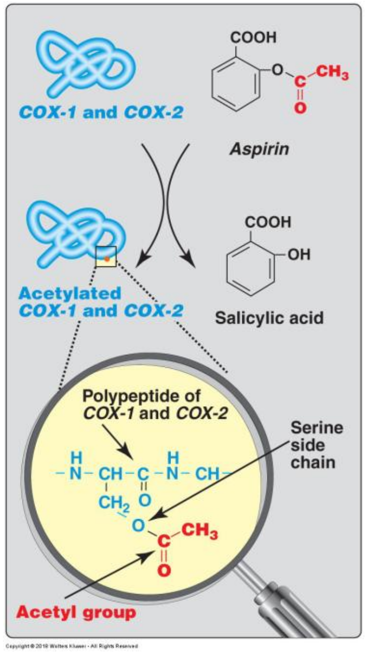
- **Cortisol** inhibits **Phospholipase A2**: Inhibits release of arachidonic acid from membrane phospholipids; Cortisol also inhibits **COX-2**; Anti-inflammatory agent
- **Aspirin** and non-steroidal anti-inflammatory drugs (**NSAID**) inhibit COX-1 and COX-2
- Selective **COX-2 inhibitor**: Celecoxib



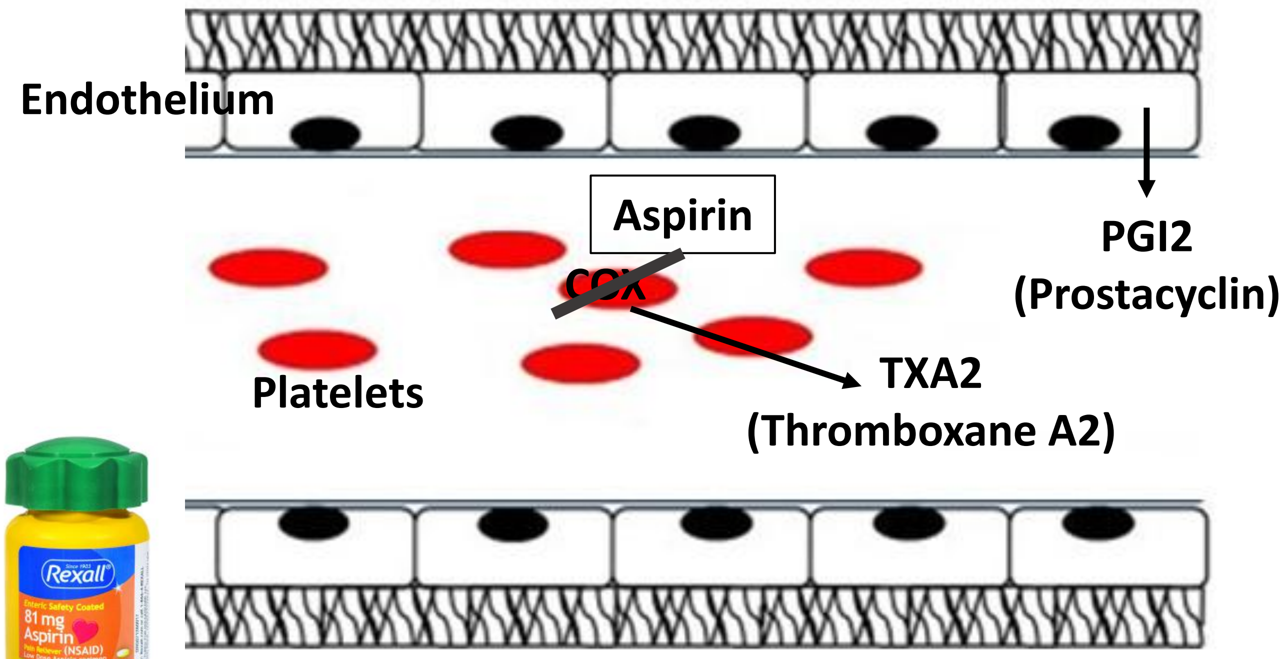




# Aspirin (Anti-thrombogenic agent)




- Aspirin irreversible inhibitor of COX
- Inhibits platelet COX (platelets cannot synthesize new enzyme as they do NOT have nucleus) less TXA2 synthesis
- Endothelium can make new enzyme as it has a nucleus; PGI2 synthesis not affected
- **PGI2** >> TXA2 inhibits platelet aggregation




# Lipoxygenase pathway: Leukotrienes

Membrane Phospholipid

| GLYCEROL | |
|---|---|
| | Saturated Fatty acid$_1$ |
| | **Arachidonic acid** |
| | **Phosphate — Nitrogen containing base** |

**Phospholipase $A_2$** ↓

**Arachidonic acid (20C fatty acid)**

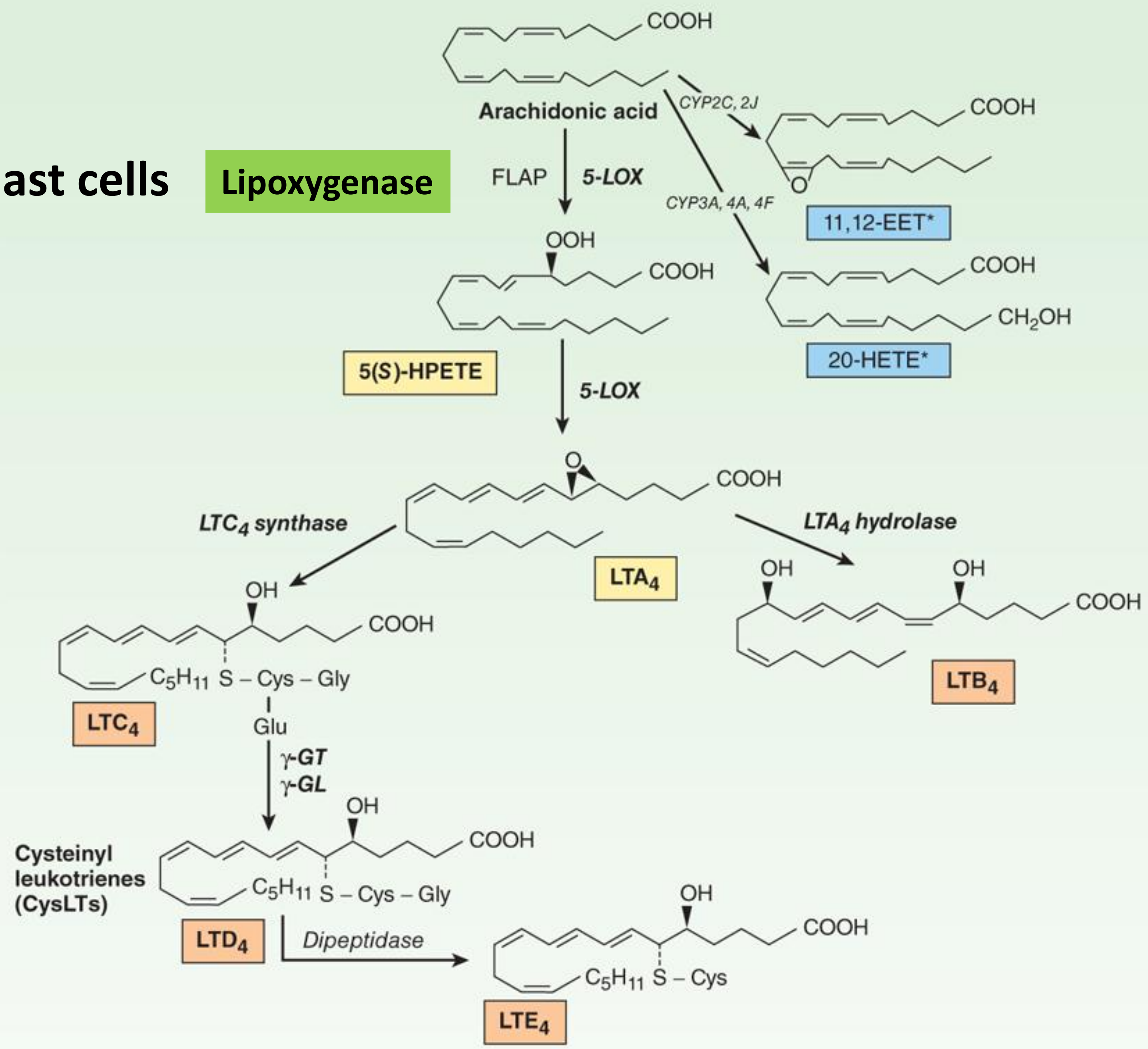
**Cyclooxygenase** → Prostanoids → PGE2, PGF2$\alpha$, PGI2, TXA2

**Lipoxygenase** → Leukotrienes LTA4 → LTC4, LTD4, LTE4; LTB4

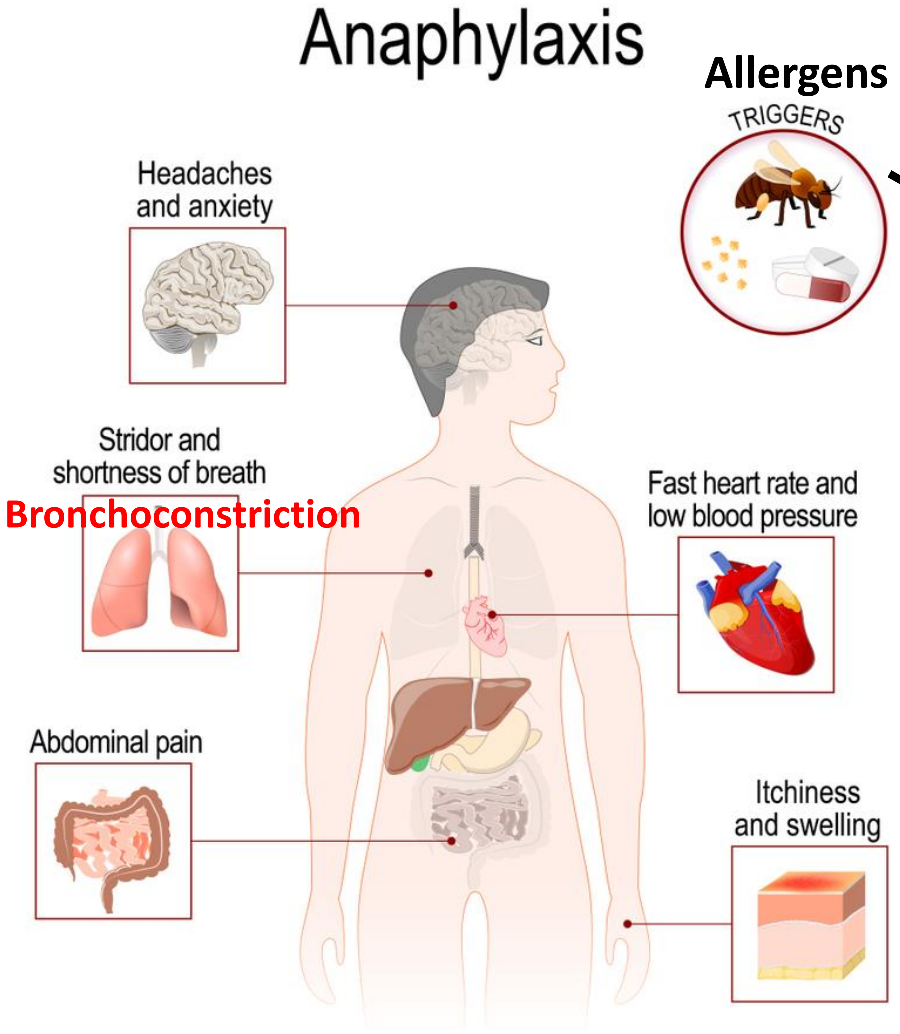
# Leukotriene synthesis

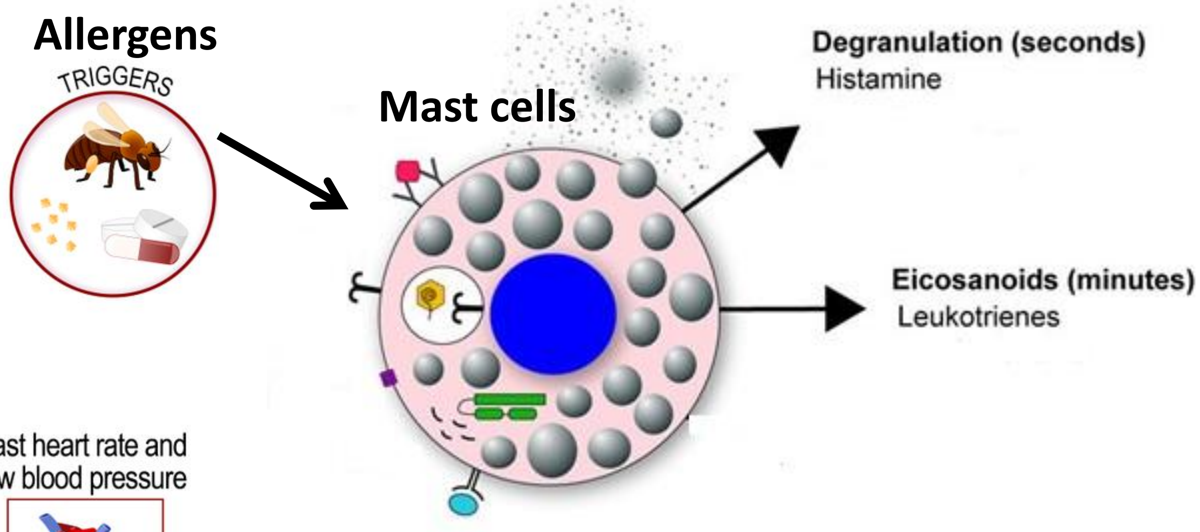
Mast cells

Lipoxygenase



# Biologic effects of leukotrienes




**Cysteinyl leukotrienes (LTC4, LTD4, LTE4)**

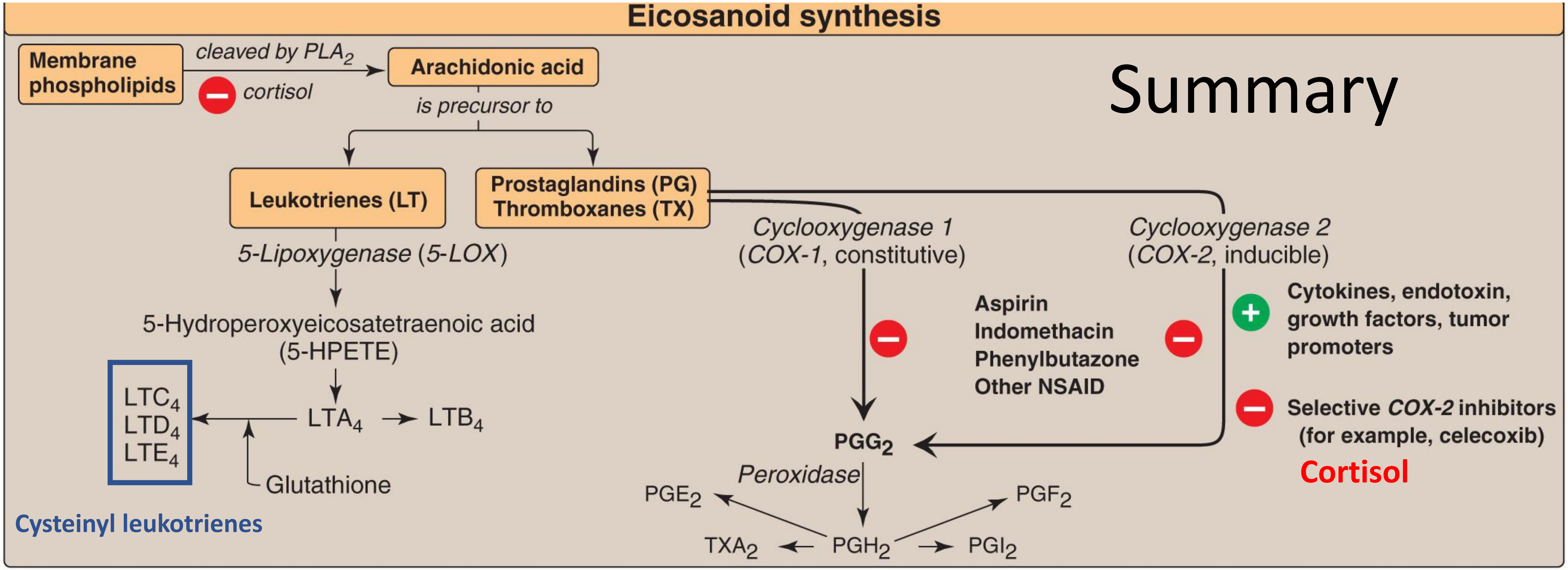
**SRS-A**

- Cysteinyl leukotrienes **(LTC4, LTD4, LTE4)**
- Released by mast cells
- Mediate allergic and anaphylactic response
- Bronchoconstriction (asthma)
- Components of **SRS-A** (Slow Reacting Substance of Anaphylaxis)

# Inhibitors of Leukotriene synthesis

- **Cortisol**: Inhibits Phospholipase A2, decreases formation of arachidonic acid and inhibits synthesis of both prostanoids and leukotrienes; Used as anti-inflammatory and anti-allergic agent
- Lipoxygenase inhibitors reduce formation of leukotrienes; Used as **anti-allergic agents (management of asthma)**

# Summary






- Phospholipase A2 forms Arachidonic acid
- COX 1 and 2 form Prostanoids (PGE2; PGF2$\alpha$; PGI2; TXA2)
- Lipoxygenase form leukotrienes: Cysteinyl leukotrienes (LTC4, LTD4, LTE4)
- Inhibitors: Cortisol; COX-1 and COX-2 inhibitors and Lipoxygenase inhibitors
- Eicosapentaenoic acid is a precursor of series-3 prostanoids

# Summary

| Eicosanoid | Major site of synthesis | Main biological activities |
|---|---|---|
| $PGE_2$ | Activated macrophages | Major mediator of inflammation; Vasodilation (redness); Pain; Swelling |
| $PGF_{2\alpha}$ | Uterine tissues | Stimulates uterine contractions |
| $PGI_2$ (Prostacyclin) | Endothelium of blood vessels | Inhibits platelet aggregation; Increases formation of cAMP in platelets; Vasodilation |
| $TXA_2$ (Thromboxane $A_2$) | Platelets | Promotes platelet aggregation; Vasoconstriction; decreases formation of cAMP in platelets; mobilizes intracellular $Ca^{2+}$ |
| Cysteinyl leukotrienes $LTC_4 \rightarrow LTD_4 \rightarrow LTE_4$ | Mast cells | Bronchoconstriction; Increases vascular permeability; Component of slow-reacting substance of anaphylaxis (SRS-A) |

# Thank you!!

Please email shupadhya@sgu.edu or ASK-BCHM@sgu.edu if you have questions